

Challenge

Convey openpilot's driving confidence states to the user while engaged.

Framing the challenge

Goals

Driver has an accurate idea of the system confidence level.

Driver is more ready to take over control at medium/low confidence level.

Driver doesn't feel the need to disengage needlessly.

Driver takes control at very low confidence levels.

Drive is still chill.

Questions

How can the existing UI and UX be improved?

How intrusive/non-intrusive should the system confidence communication be?

At what confidence level would the driver take over?

How granularly should the confidence level be communicated? 1-3, 1-5, 0-100, analog?

Is it valuable for the driver to know detailed limits; steering limit, break limit, acceleration limit?

Car interface design principles

Glancability

Information should be quickly understandable.

Avoid distractions

Only communicate when needed and at an appropriate level of intrusiveness.

Talk like a human

Behave and communicate like a human driver rather than a robot.

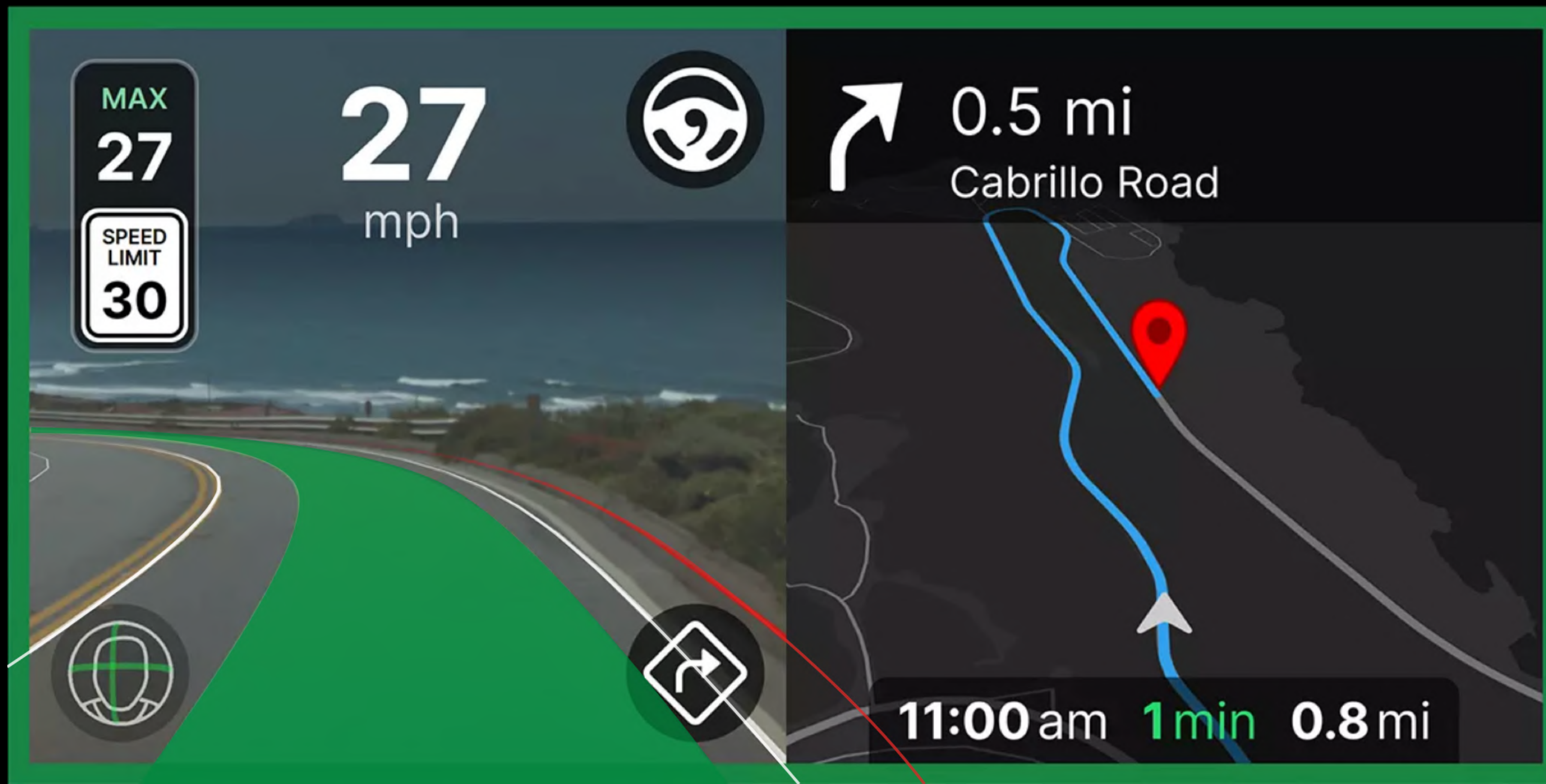
Be concise

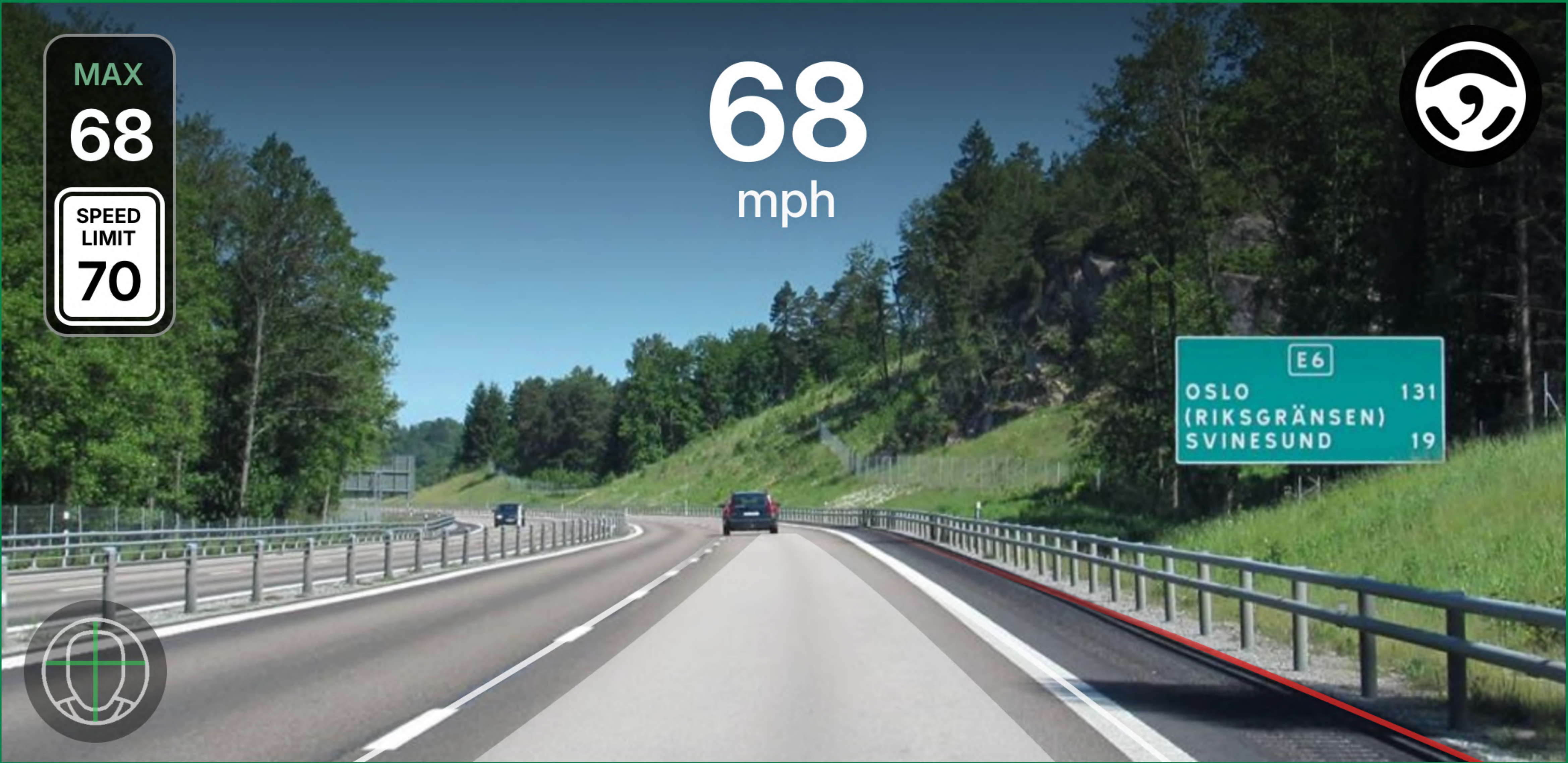
Writing should be short and the most important information should come first.

Hide complexity

Making the UI a literal reflection of the underlying logic/implementation is rarely helpful

How can the existing UI and UX be improved?





Engaged



68
MPH

SPEED
LIMIT
70

E6
OSLO 131
(RIKSGRÄNSEN)
SVINESUND 19



Not engaged



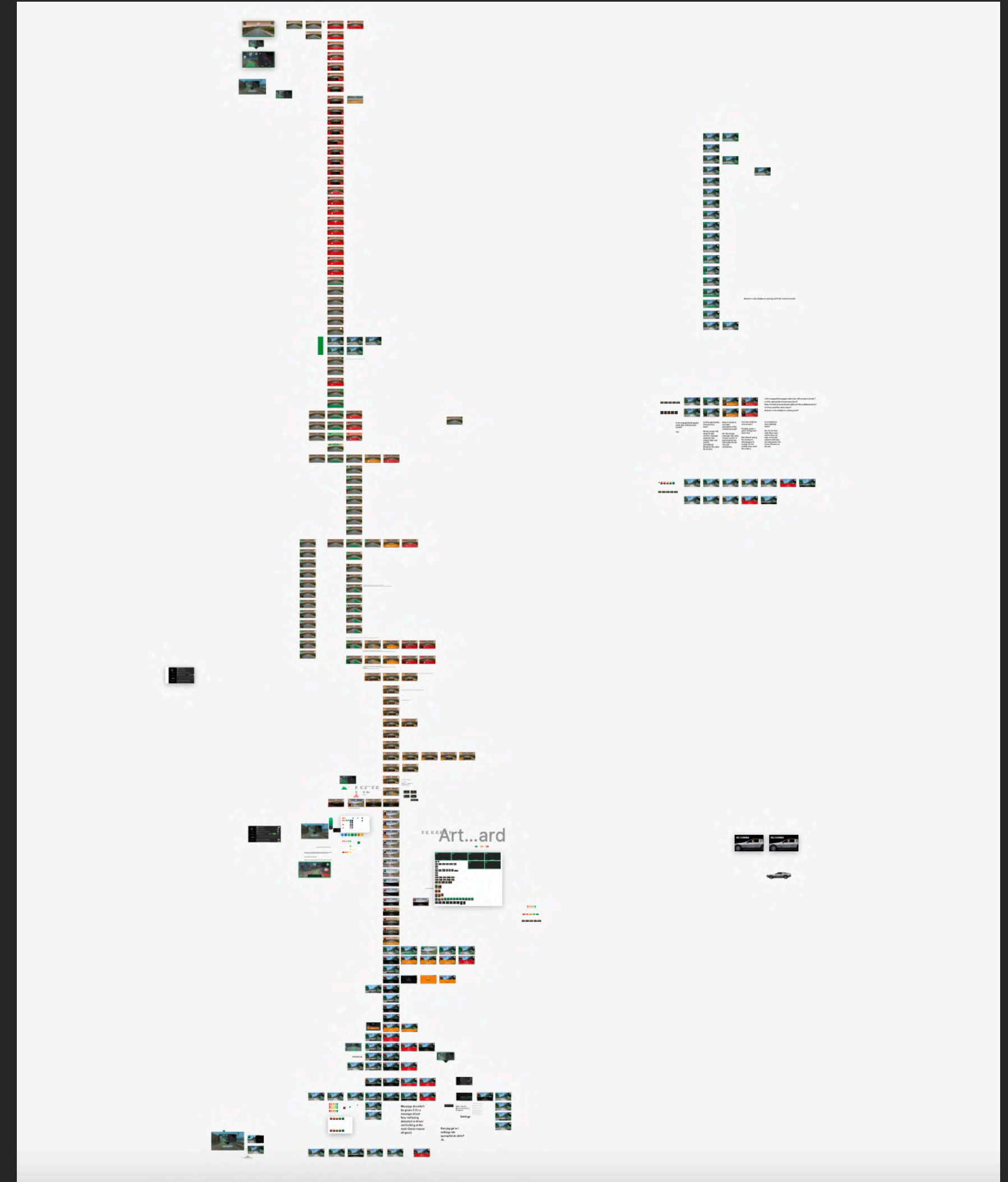
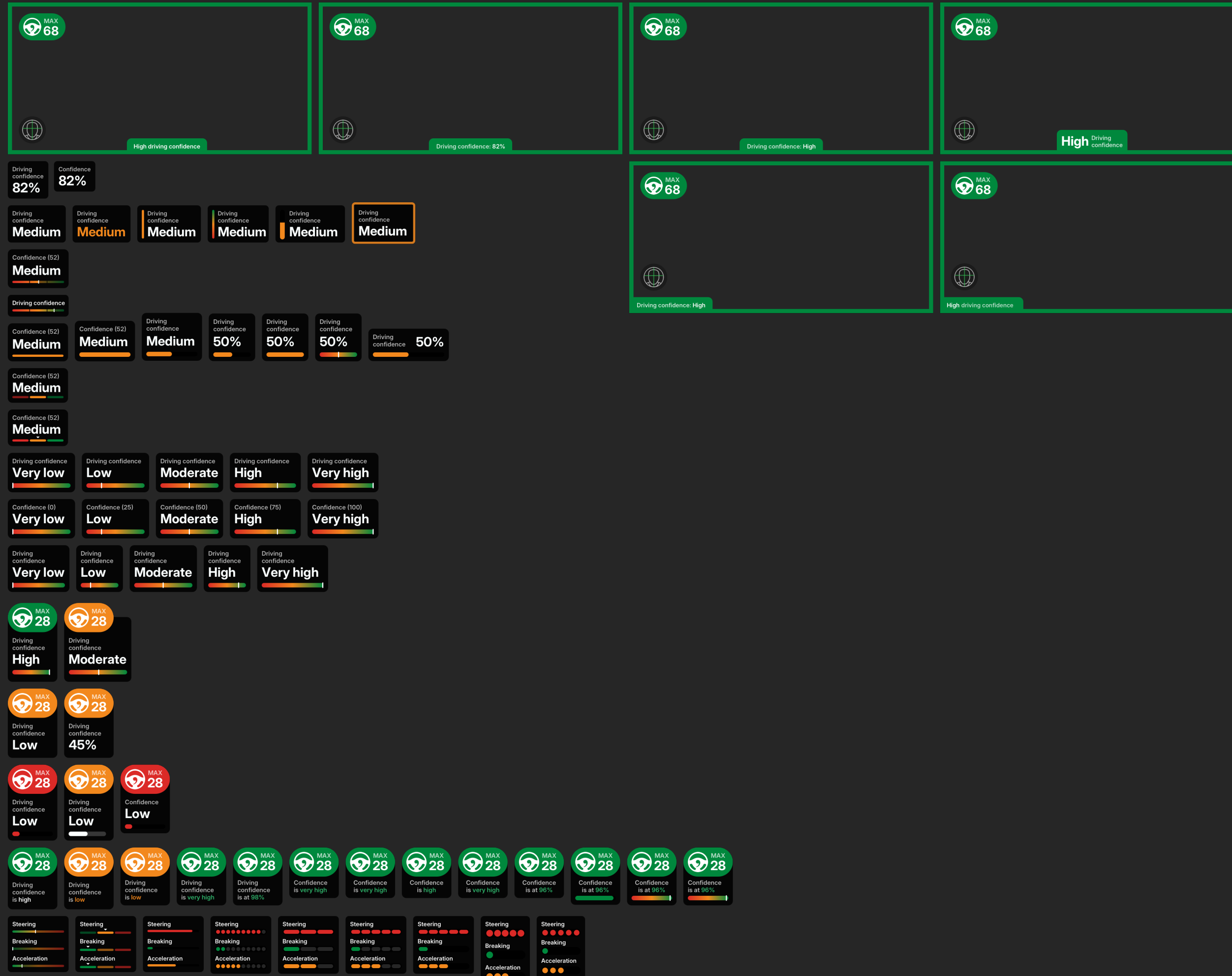
68
MPH

SPEED
LIMIT
70

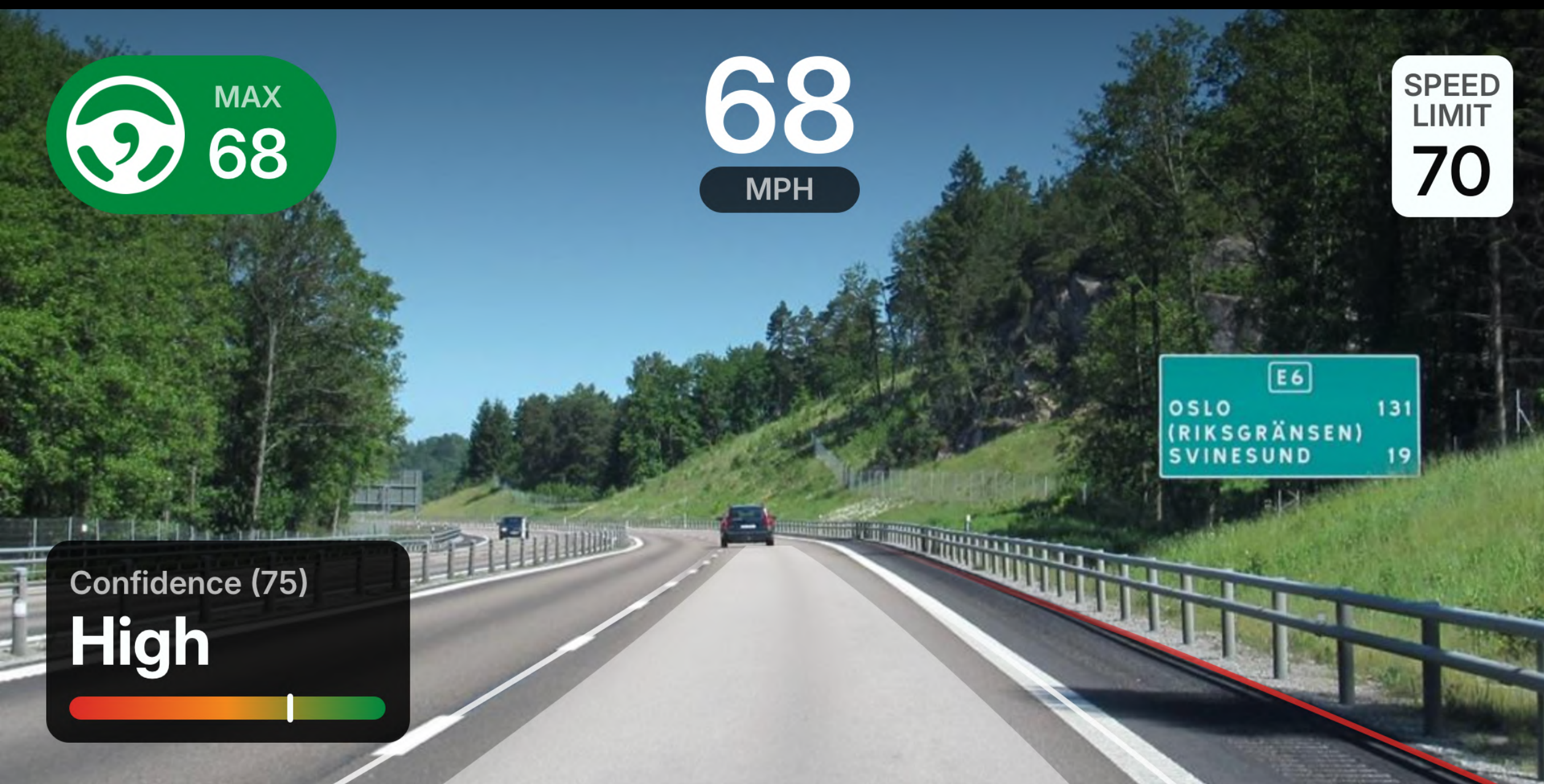


Engaged

Exploration



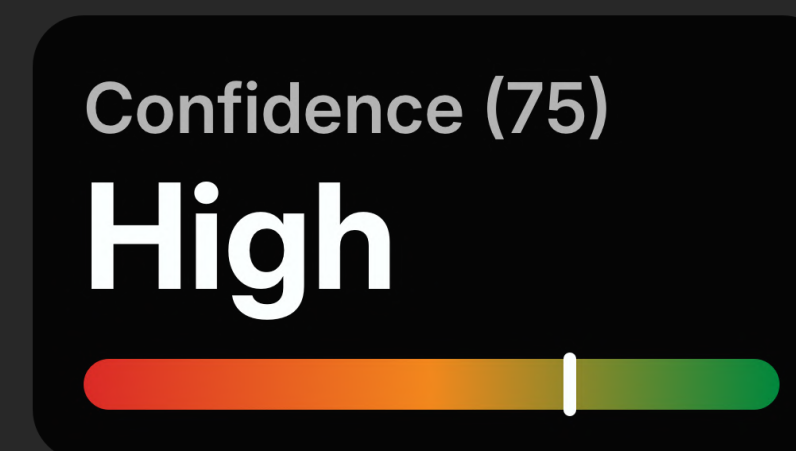
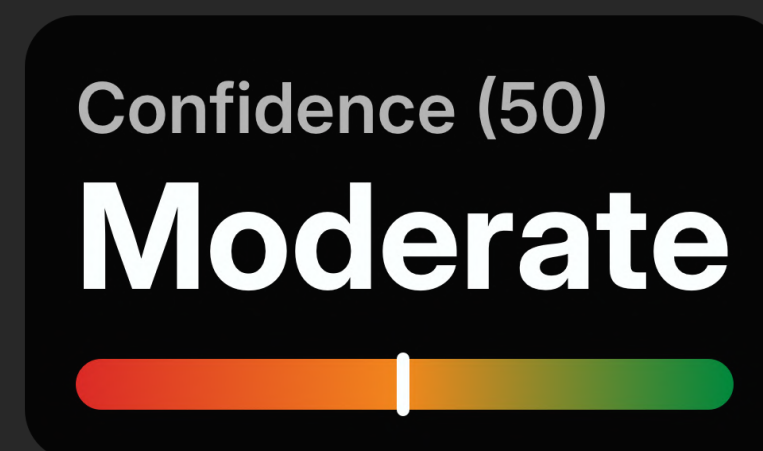
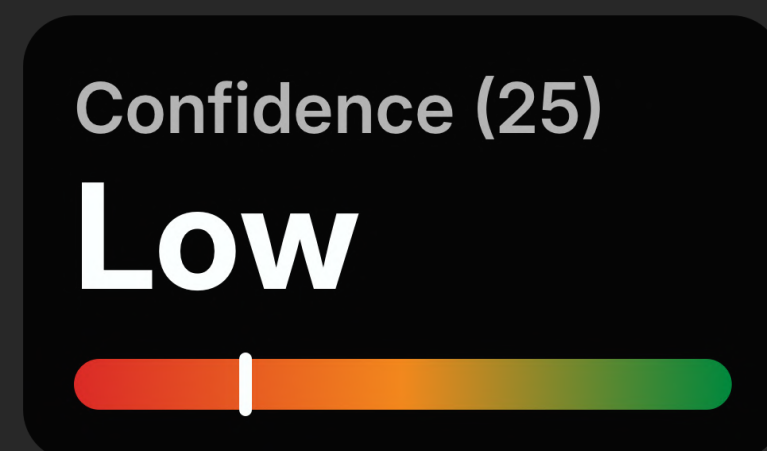
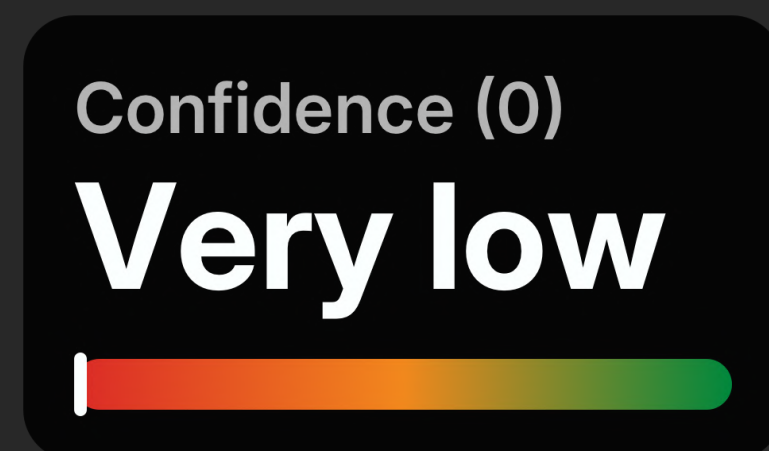
Worded versus limits



Worded



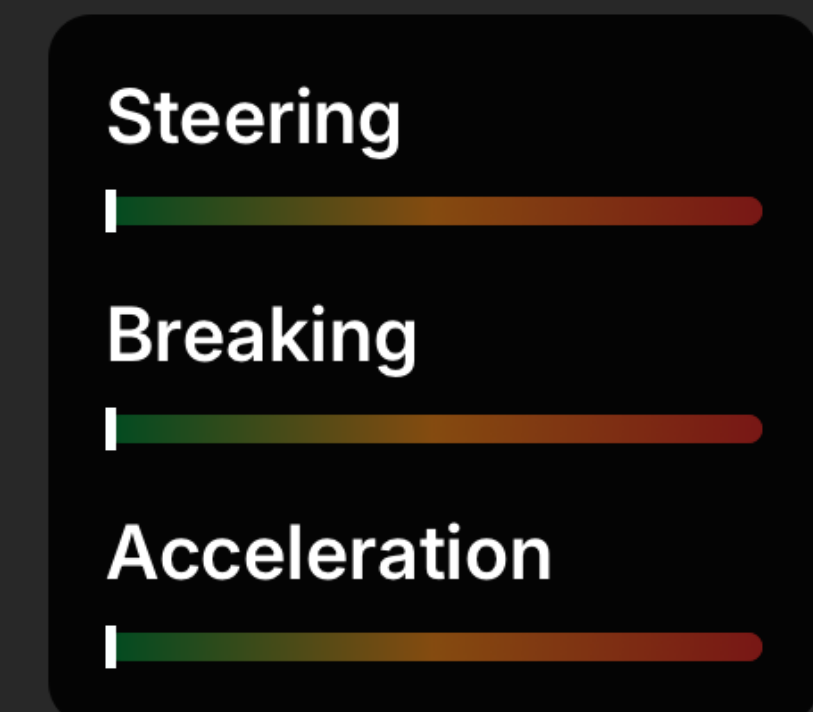
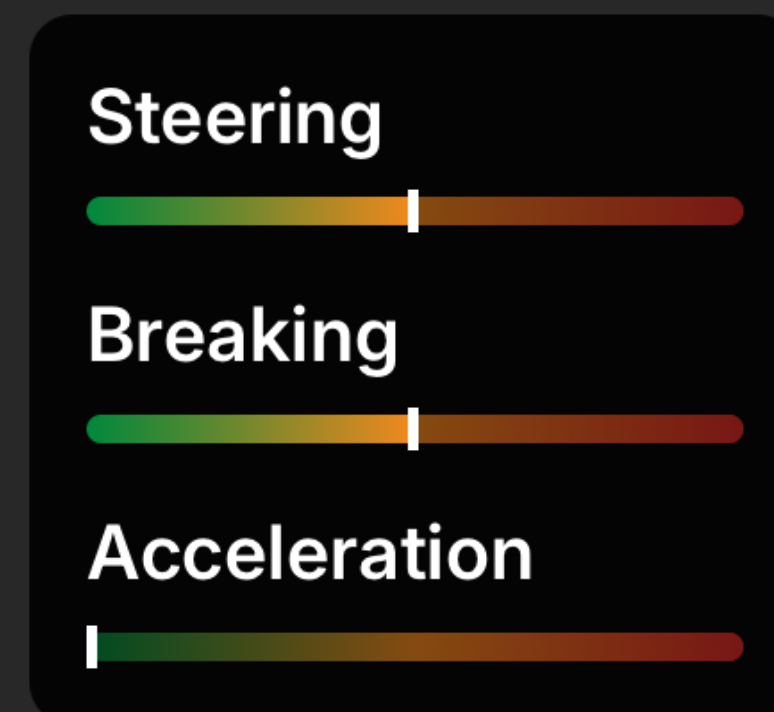
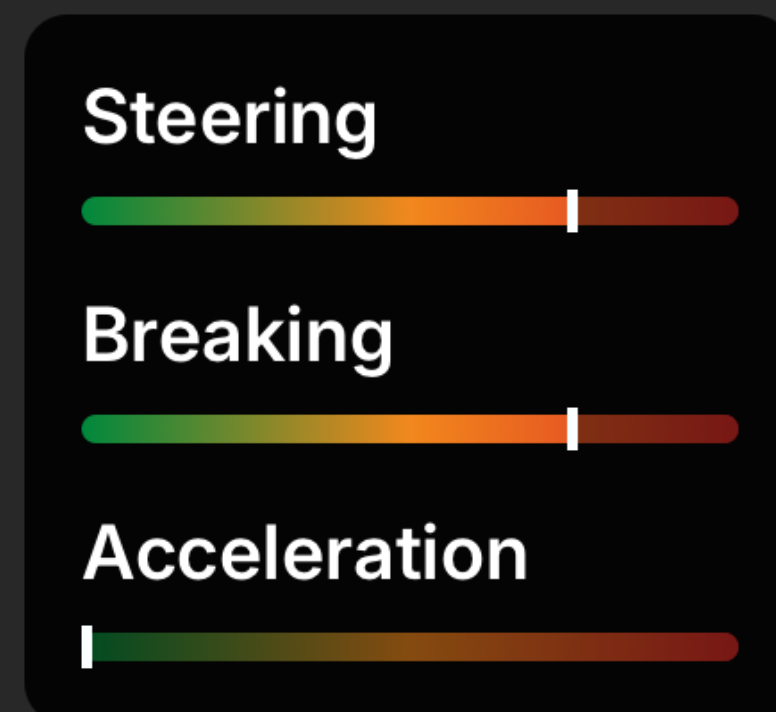
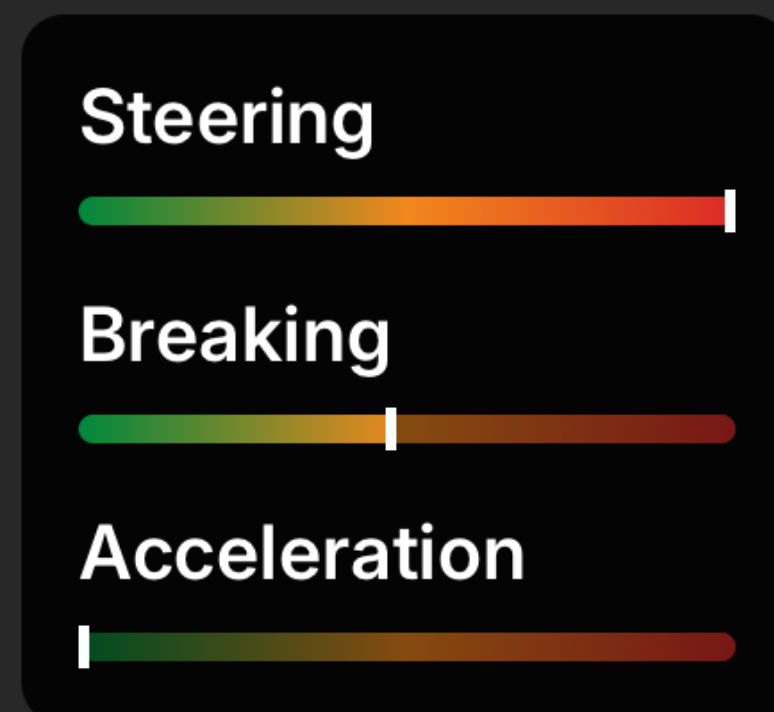
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Limits



+



User test round 1

Questions

How intrusive/non-intrusive should the system confidence communication be?

How granular should the confidence level be? 1-3, 1-5, 0-100, analog?

Are steering, breaking and acceleration limits helpful to see?

Is the colorcoded confidence level clear and helpful?

At what confidence level would the driver take over?

The user



Gender
Woman

Age
33

Occupation
Manager

Country
Sweden

Openpilot usage
Daily user (not buyer/installer)

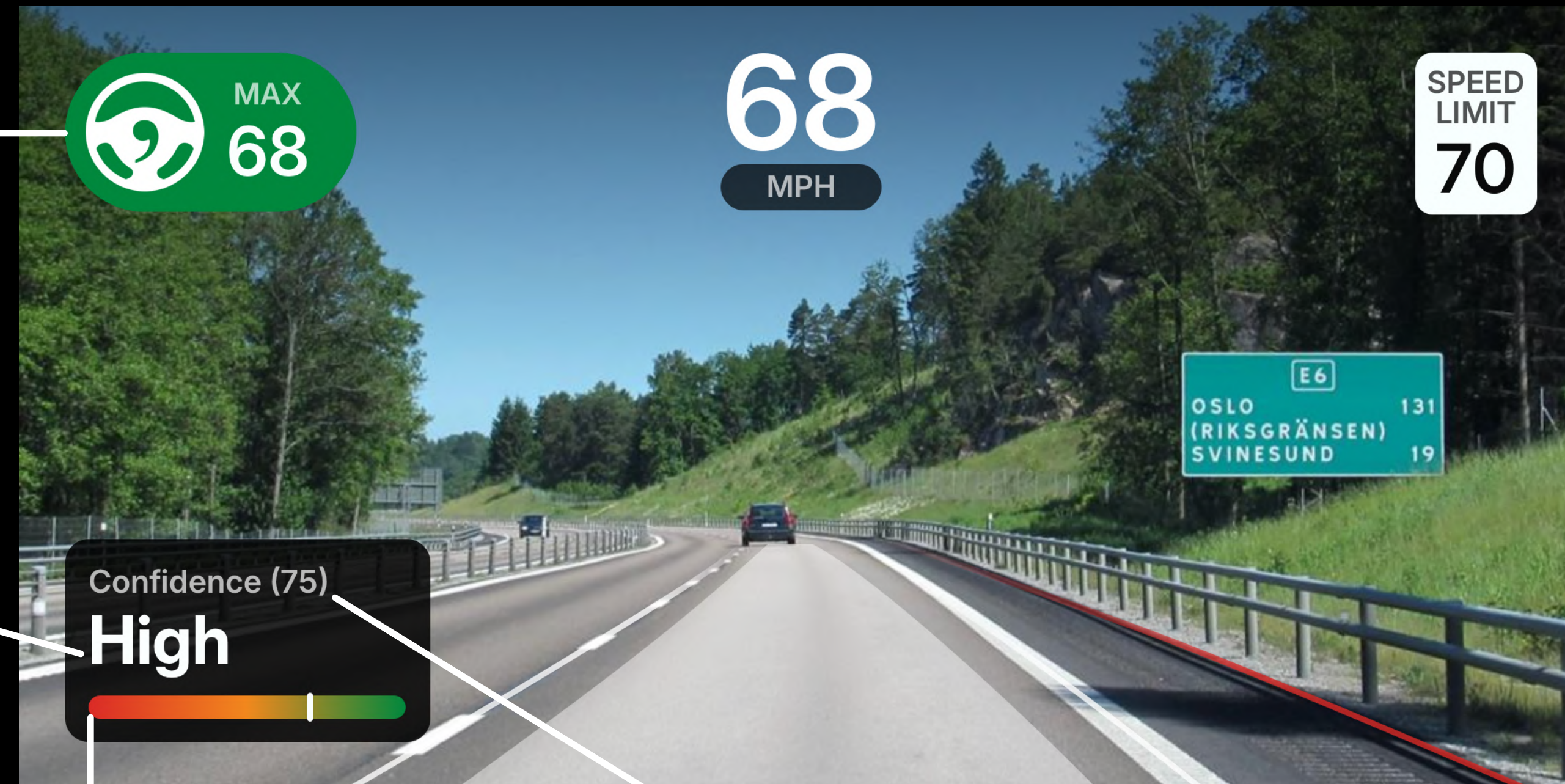
Technical level
Moderately savvy



Worded



"Green means good,
that I don't have to do
anything."



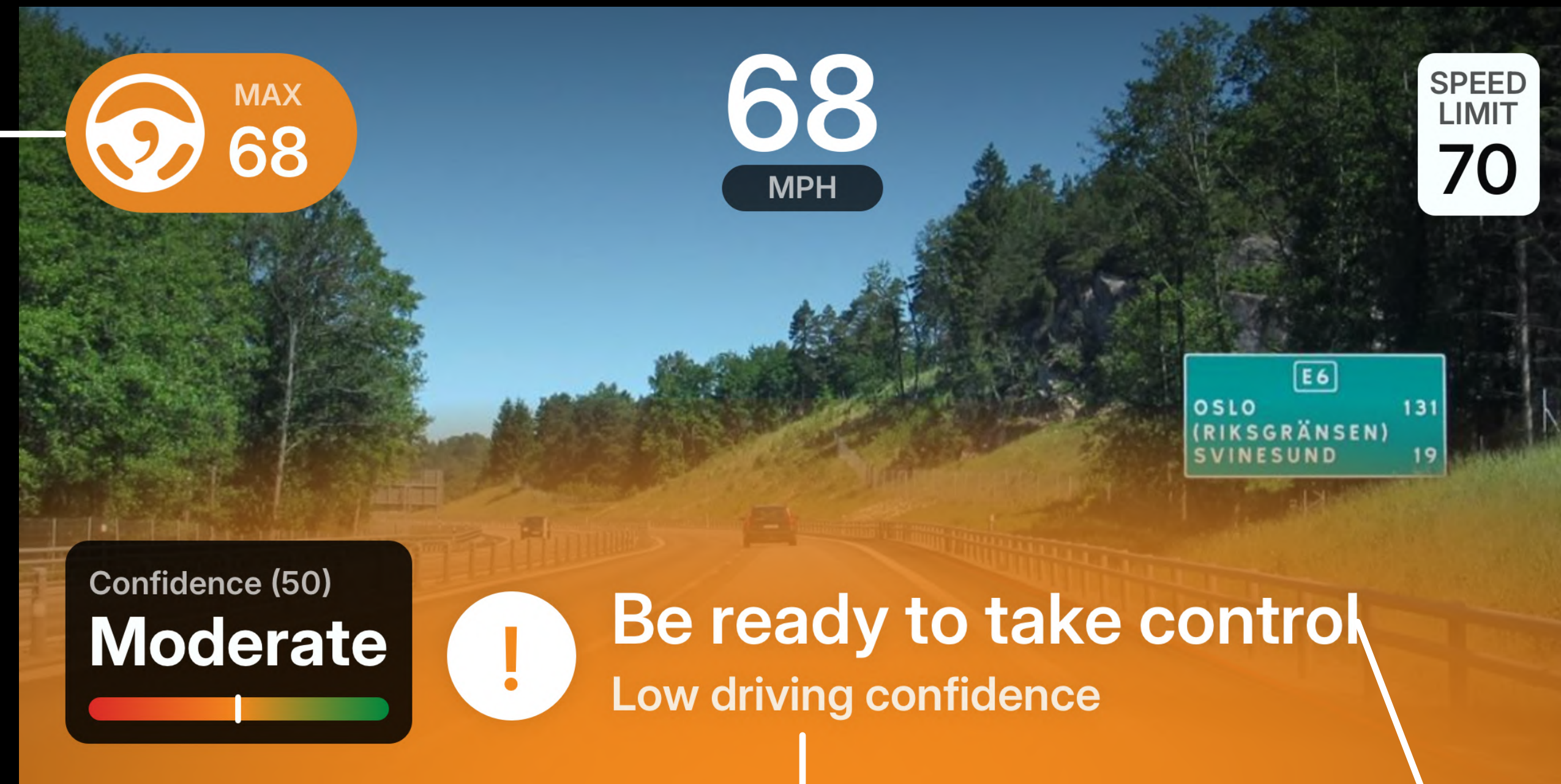
"I think it means how
confident it is to drive,
that the road is good
enough."

"I think it will turn off
when it's down on
20-30, in the red."

"I don't think I really
need the number."

"I think the color is how confident it is at driving, or it might turn orange if I go above the speed limit."

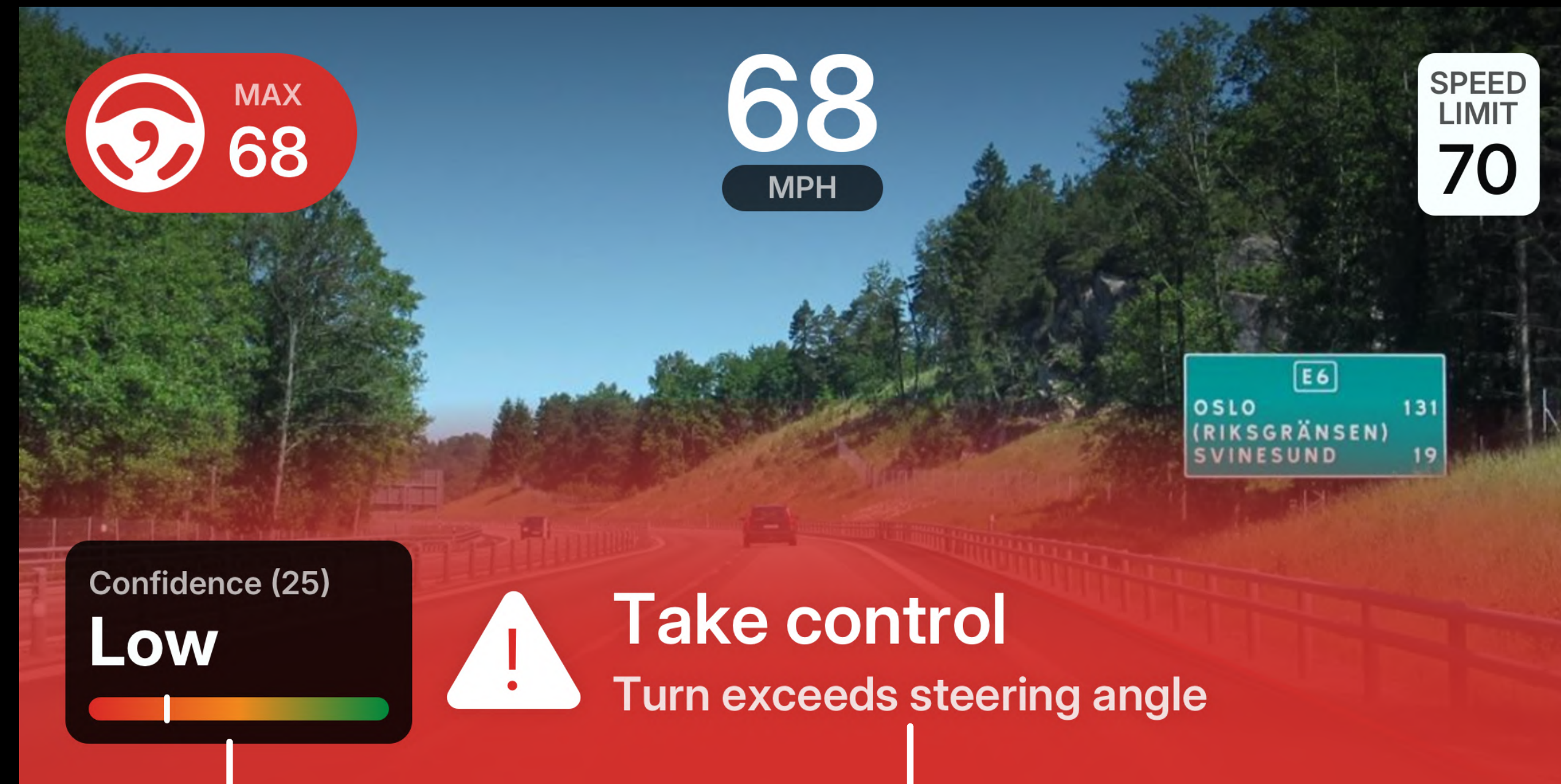
"If I saw this I would have taken the wheel and turned it off by pressing the gas pedal or breaking."



"I'm confused by it saying 'Low driving confidence' but on the left it says 'Moderate'."

"'Be ready', it wants me to take control. I think it will turn off eventually if I don't take control."

"As soon as it's red I'd take over. I'm thinking it will turn off if it doesn't feel any gas or breaking."



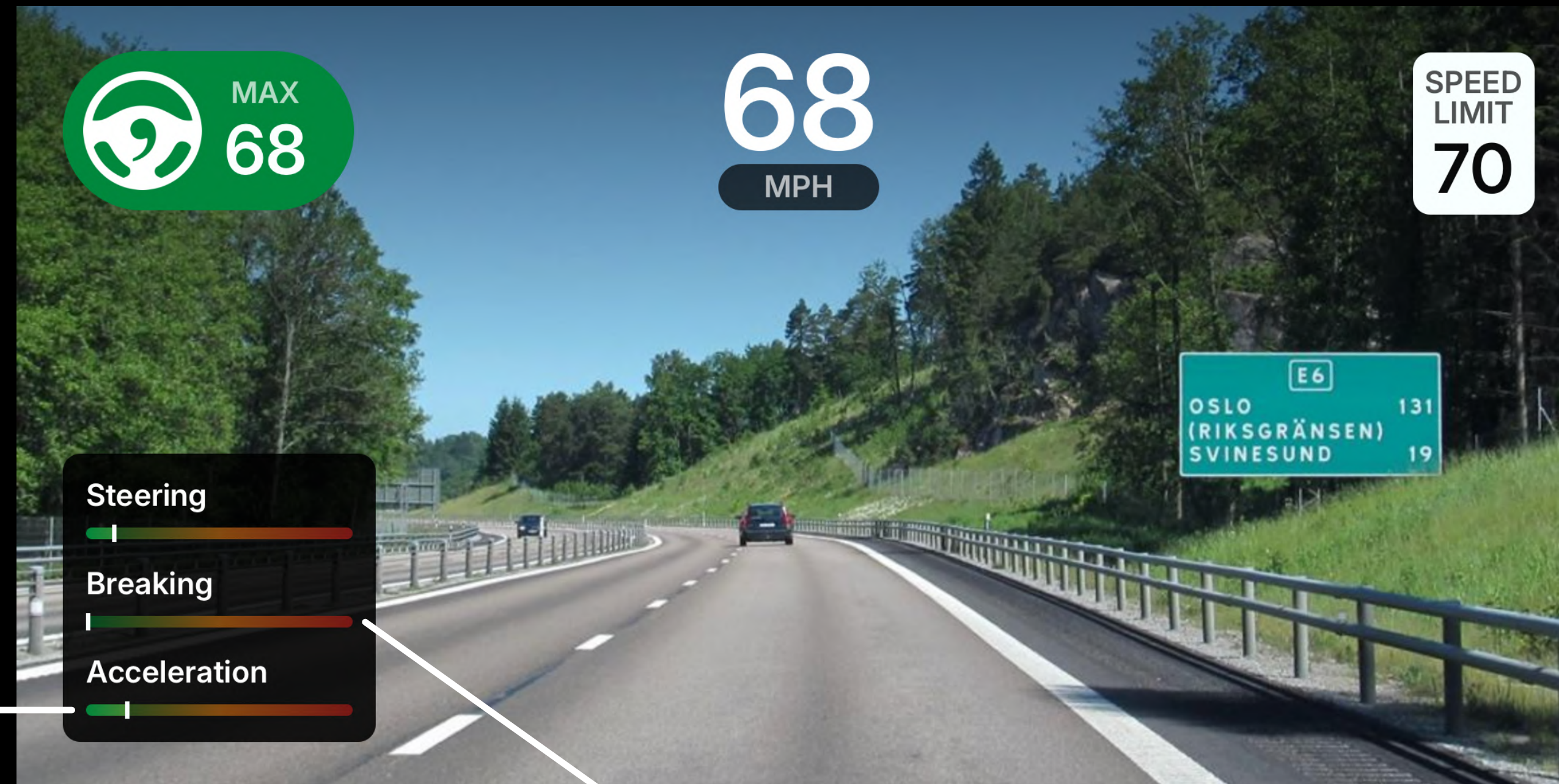
"When I take control, confidence will disappear."

"I guess it means that the turn is too sharp."

Limits



"I don't understand anything."



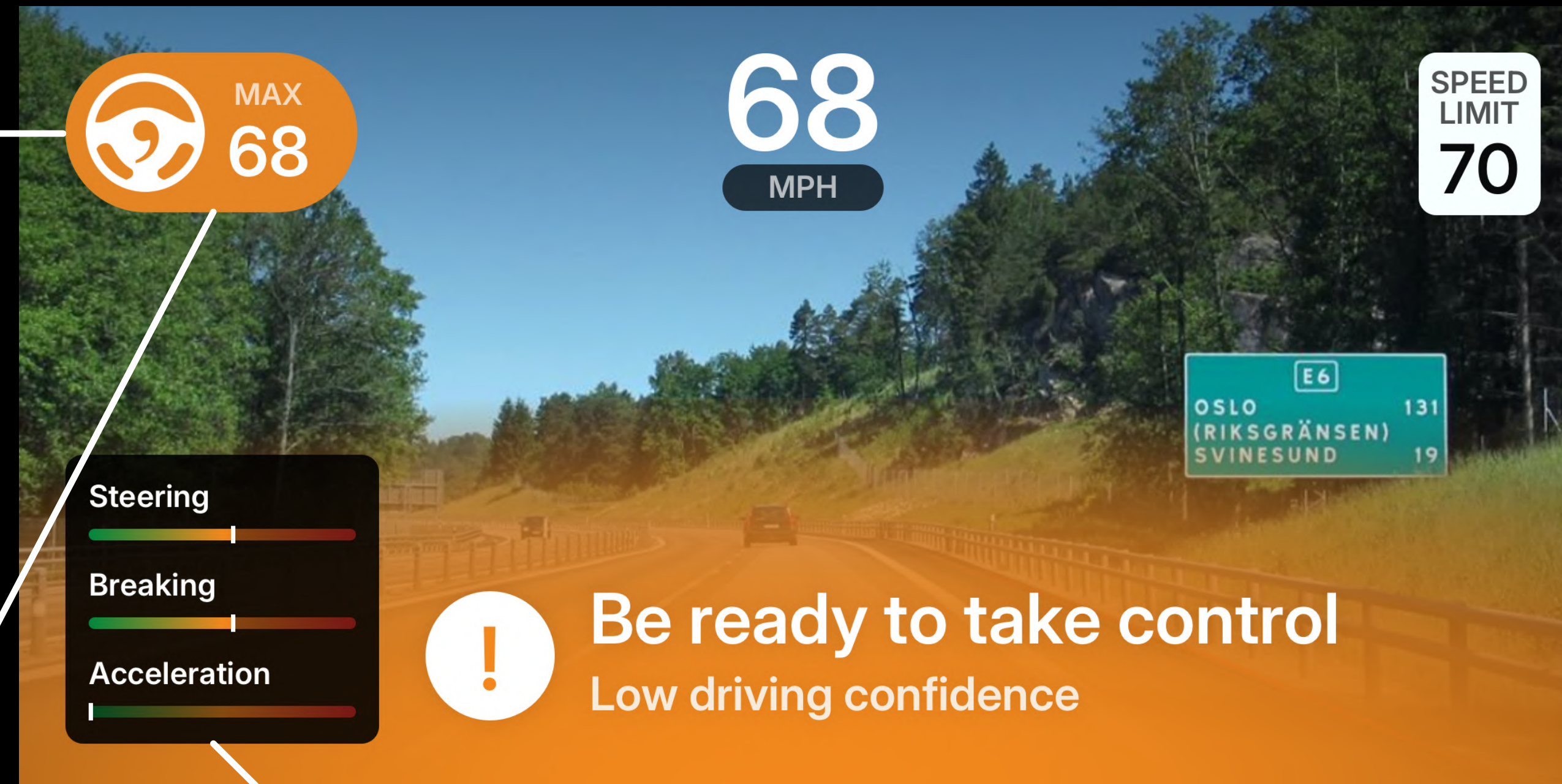
"Does acceleration turn red if there's a car in the way? But if there's a car in the way then it's a good thing that it doesn't accelerate so then it should be green?"

"Breaking feels binary, either you break or you don't."

"I just want it to be simple, is it good or bad? Do I need to drive manually or not?"

"I want the color to be confidence, rather than if I go above the speed limit like in Google Maps. I already know what the speed limit is."

"I would have hands more ready on the wheel and the foot more ready on the break if it turned orange or red."



"I don't need to know the technical stuff, I just want to know if it can drive."

Ustertest learnings

The user just wants to know if they should/are able to; 1. drive manually 2. not drive manually 3. be more ready to take control.

Steering, breaking and acceleration limits are confusing and too technical for the user.

Not entirely clear that the color is connected to confidence level.

The user will disengage even though they don't need to if the communication at moderate confidence levels is too strong/intrusive.

Engaged/not engaged state is clear for the user even without screen border.

Worded + combined





Not engaged



Engaged

User test round 2

Questions

Is the confidence level
communicated in a way that
doesn't make the user disengage
needlessly?

Is it clear that the color is
connected to the confidence
level?

Is more conversational writing
valuable?

"I like that it's connected to the speed, it has confidence that it can drive at this speed."

"I like 'Confidence is..', a bit more like something you'd say."

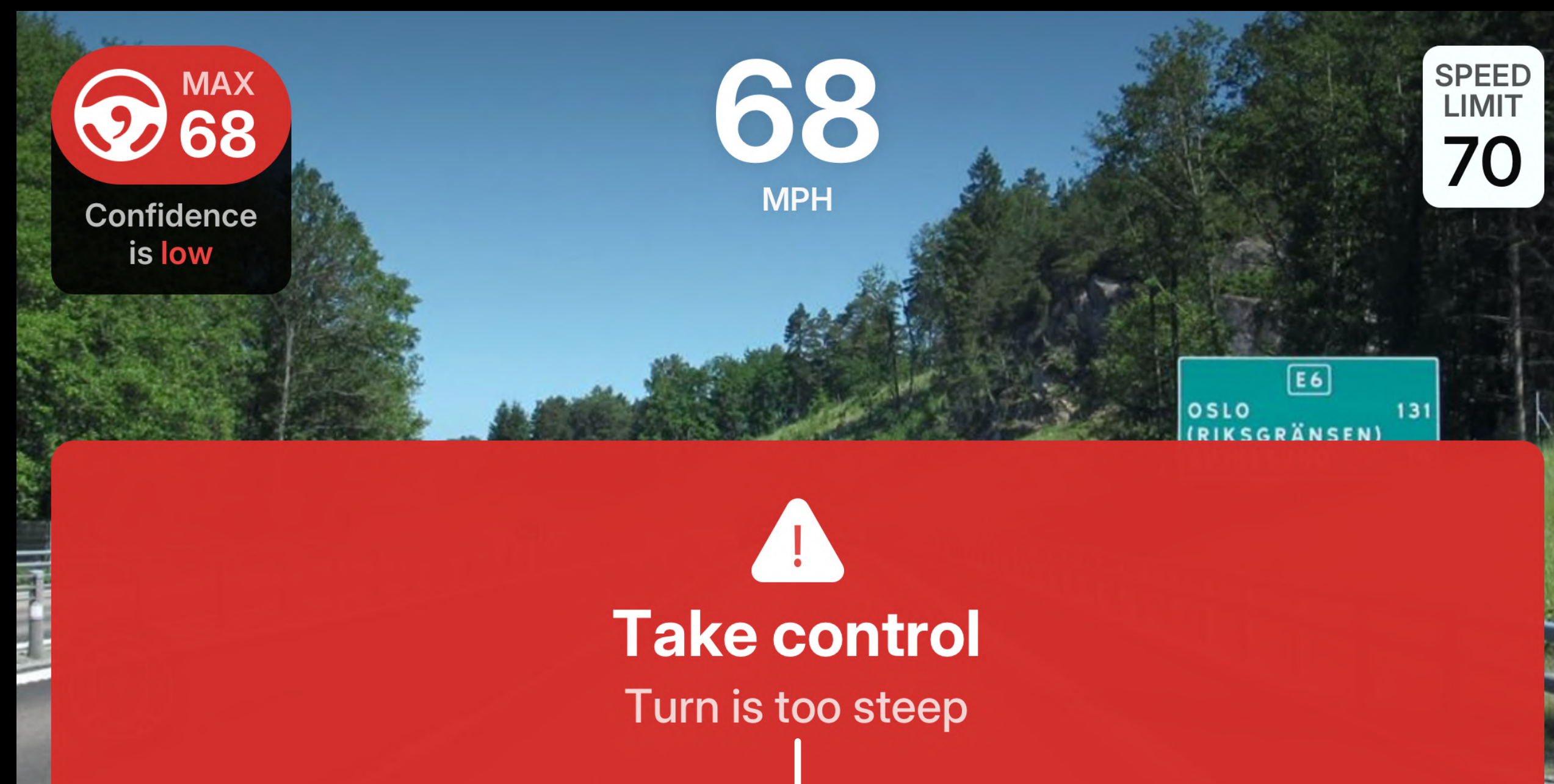


"I would have been more alert but I wouldn't have turned it off. "

"A ping would have made me look and put my hands on the wheel and be more ready."



"I would have turned it off right away."



"I don't need to know why, I just need to know that I should take over. "

Usertest learnings round 2

The user prefers 3 over 5 confidence levels.

Color connection to confidence level is now clear.

A subtle “ding” sound might be helpful when stepping down to moderate confidence level to increase readiness of taking over.

The user appreciate the conversational writing.

The user would no longer disengage needlessly at moderate confidence levels.

Final design



68
MPH

SPEED
LIMIT
70



E6
OSLO 131
(RIKSGRÄNSEN)
SVINESUND 19



Confidence
is **high**

68
MPH

SPEED
LIMIT
70

E6
OSLO 131
(RIKSGRÄNSEN)
SVINESUND 19





Confidence
is **high**

68
MPH

SPEED
LIMIT
70



Watch the road



Confidence
is **moderate**

68
MPH

SPEED
LIMIT
70

E6
OSLO 131
(RIKSGRÄNSEN)
SVINESUND 19





Confidence
is **low**

68
MPH

SPEED
LIMIT
70

E6
OSLO 131
(RIKSGRÄNSEN)



Take control
Turn is too steep

Next steps

User test: Ideally 8-12 people,
minimum 5.

Community feedback.

Plan for measuring after shipping.

Questions

Is a few, 3 or 5, confidence levels the right way to go?

Is the states glancable enough without the screen border?

Would audio feedback when stepping down confidence level be helpful?

What should the middle confidence level be called; moderate, medium, average?

Thanks!